Data Science

Interpretable Machine Learning

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Outline

1. Introduction

2. Explainability and interpretability
   - Model-agnostic interpretation methods
   - Local and global surrogate models
   - Shapley Values

Model Interpretability and variable relevance

- In a stimulating and provocative paper, Breiman (2001)[1] shook the statistical community by making it to be aware that traditional Statistics was no longer the only way to learn from data:
  - Data Modeling Culture (traditional Statistics):
    - Linear regression, logistic regression, additive models, etc.
    - They allow to interpret how the response variable is associated with the input variables: Transparent models.
  - Algorithmic Modeling Culture (Machine Learning, Data Science):
    - Neural networks, support vector machines, random forest, etc.
    - They have extremely good predictive accuracy, and they usually outperform in this criterion statistical models.
    - Low interpretability: Black boxes.
- An apparent dichotomy: predictive capacity versus interpretability.
- Breiman claimed for procedures allowing better interpretation of the algorithmic models results, without giving up their predictive ability.

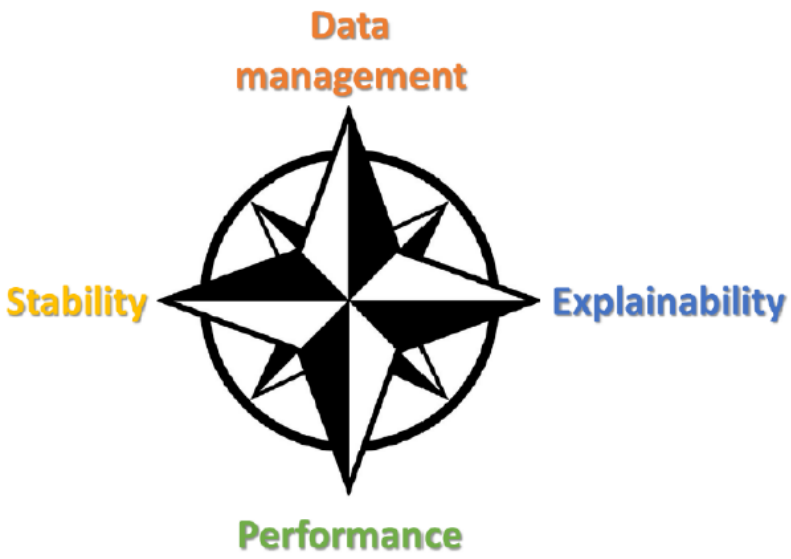
[1] Breiman, L. (2001). Statistical modeling: The two cultures. Statistical Science 16, 199-231.

Looking for interpretability is mandatory

- Machine learning community has been worried about interpretability: if the users do not trust a model or a prediction, they will not use it (Ribeiro, Singh, and Guestrin 2016)
- In 2018 the General Data Protection Regulation of the European Union established the users' right to explanation: when an algorithmic decision significantly affects a user, he or she has the right to ask for an explanation of such a decision.
- A powerful research line has been developed: Interpretable Machine Learning, eXplainable Artificial Intelligence.
  - Several review papers (Barredo-Arrieta et al. 2020 is one of the most recent and extensive reviews).
  - Three monographs: Molnar (2019), Biecek and Burzykowski (2021) and Masís (2021).

Governance of AI in Finance

ACPR (2020) suggests four evaluation principles for AI algorithms and models:



IML/XAI concepts

- Desirable properties for predictive models: transparency, interpretability, explainability.
- Not well defined concepts that are difficult to be measured.
  - Lipton 2018: The term interpretability is ill-defined.
  - Barredo-Arrieta et al. 2020: The derivation of general metrics to assess the quality of XAI approaches remain as an open challenge.

Transparency, interpretability, explainability

- An algorithm is said to be transparent if the mechanism by which it works can be understood by a human (Lipton 2018, Barredo-Arrieta et al. 2020).
- This definition encompasses different degrees of algorithm transparency, given the wide range of human expertise (Lipton 2018).
- The concept of interpretability is both important and slippery, as acknowledge by Lipton (2018), who mentions that a general goal of interpretability might simply be to get more useful information from the model.
- Barredo-Arrieta et al. (2020) consider than transparency and interpretability are synonymous in this context, and that what is relevant is what information you want to extract from a model and how to get it.
- Miller (2019) gives a slightly different sense to interpretability, equating this term to explainability with the meaning of how well a human could understand the decisions in the given context, that is, the ability of an algorithm to provide humans an explanation for any of its particular decisions.

Transparency, interpretability, explainability

- We can summarize telling that in the last years the quality perception of a prediction algorithm is no longer focused exclusively on the accuracy of predictions.
- In addition to that, the possibility of obtaining information on the performance of the algorithm, in both the global and local sense, is now appreciated.

Global versus local interpretability

- Information about the global performance refers to determining which is the role of each explanatory variable in the prediction process over the whole support of the explanatory variables.
  - Global interpretability: Measures of variable importance or relevance.
- On the other hand, the goal of understanding local performance is to provide a meaningful explanation of why the algorithm returns a certain prediction, given a particular combination of the predicting variables values.
  - Local interpretability: Why the prediction model does a particular prediction for a given individual?
- The local aspect of interpretability is directly related with the users' right to explanation advocated for by, for instance, the EU's GDPR.

Transparent models versus black-box models

According Barredo-Arrieta et al. (2020) and Maksymiuk, Gosiewska, and Biecek (2020), the prediction models can be classified as follows:

1. Transparent models, or interpretable by design models, or white-boxes, or glass-boxes.
   - Models that, by design, have an easily interpretable structure.
   - Linear models (LM, and generalized linear models, GLM), generalized additive models (GAM, including additive models), classification and regression trees (CART), decision rules, k-nearest neighbors, and Bayesian models (including Naive Bayes prediction rules).
   - They offer sufficient interpretation and/or diagnostic tools, both numeric and graphic.
2. Non-transparent models or black-box models.
   - Their design does not provide a directly interpretable structure.
   - These models require additional interpretation tools.
   - All the prediction methods not explicitly mentioned before.

Non-transparent models

Non-transparent models can be divided into two subgroups:

1. Models for which there exist model-specific methods for knowledge extraction. These methods require full access to the model structure:
   - Tree ensembles (including random forests and boosted methods).
   - Neural networks (NN, including deep learning based on multi-layer, recurrent or convolutional NN).
   - Support vector machines (SVM).
2. The rest of the models, only model-agnostic methods are available for interpretation:
   - Do not need to know the internal structure of the prediction model to be explained.
   - Only requirement: the ability to evaluate the prediction model repeated times on data from the training or the test set, or on perturbations of them.
   - They can be applied to any predictive model, even to those having model-specific methods or those that are transparent models.

Non-transparent models

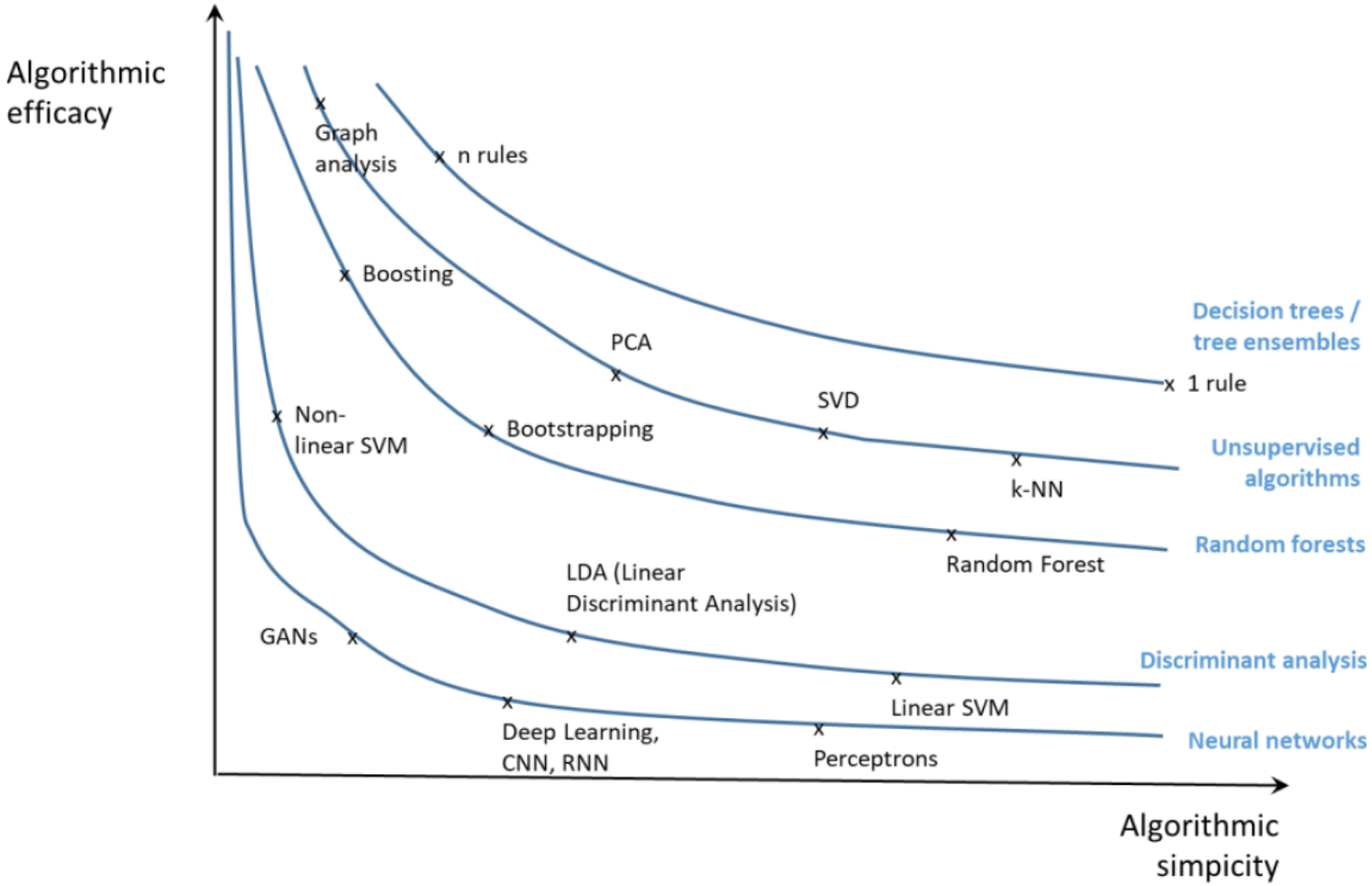
- All the interpretation methods applicable to non-transparent prediction models are globally known as post-hoc interpretation methods, a term that encompasses model-specific as well as model-agnostic methods.
- The results provided by these methods can be numerical and graphical, although most of the methods choose one or the other format.
- Finally, it is worth mentioning that most of the interpretation methods are heuristic, and only some of them are derived from a formal axiomatic statement.

Classification of models and interpretability tools

Transparent models	Black-boxes: Post-modeling interpretability
Linear model (LM) GLM GAM CART Rule based models Naïve Bayes k-nearest neighbours	**Model-specific methods:** • Tree ensembles • Neural networks • Support vector machines **Model-agnostic methods:** Global measures • Variable importance by - Leave-one-covariate-out (LOCO) - Perturbing a variable in the test set: Random permutations, knockoffs, **Ghost-variables**, ... • Variable importance based on Shapley's value • Partial dependence plot (PDP) • Accumulated local effects plot (ALE) Local measures • Local interpretable model-agnostic explanations (LIME) • Local variable importance based on Shapley's value • SHAP (SHapley Additive exPlanations) • Break-down plots • Individual conditional expectation (ICE) plot, or ceteris paribus plot

Simplicity/efficacy tradeoff of ML algorithms

Source ACPR (2020):



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The main five interpretation methods

1. Partial Dependence Plot (PDP)
2. Individual Conditional Expectation (ICE)
3. Accumulated Local Effects (ALE)
4. Local interpretable model-agnostic explanations (LIME)
5. SHapley Additive exPlanations (SHAP)

Partial Dependence Plot (PDP)

Definition

The Partial Dependence Plot (PDP) shows the marginal effect one or two features have on the predicted outcome of a machine learning model (Friedman 2001).

Partial Dependence Plot

The Partial Dependence function $pd_s(.)$ is the expectation of the model output $\hat{f}(.)$ over the marginal distribution of all variables other than Xs.

$$pd_s(x_s) = \mathrm{E}_{X_c}\left(\hat{f}(x_s, X_c)\right) = \int \hat{f}(x_s, x_c) d\mathcal{P}(x_c) \quad \forall x_s \in \Omega_s$$

Partial Dependence Plot (PDP)

PD estimate

In practice, PD function is simply estimated by averaging over the training data as:

$$\hat{pd}_s(x_s) = \frac{1}{n}\hat{f}(x_s, x_c^{(i)}) = \int \hat{f}(x_s, x_c) d\mathcal{P}(x_c) \quad \forall x_s \in \Omega_s$$

Interpretation

The Partial Dependence Plot (PDP) displays the values of the estimated PD function $\hat{pd}_s(x_s)$ for some selected values x_s of the feature X_s and shows whether the relationship between the target and a feature is linear, monotonic or more complex.

Advantages

1. Intuitive: The partial dependence function at a particular feature value represents the average prediction if we force all data points to assume that feature value.
2. Interpretation is clear: If the feature for which you computed the PDP is not correlated with the other features, the PDP plot shows how the average prediction in your dataset changes when the sth feature is changed.
3. Easy to implement: the PDP does not require re-estimation or re-training of the model.
4. Causal interpretation: We intervene on a feature and measure the changes in the predictions. In doing so, we analyze the causal relationship between the feature and the prediction in the model (Zhao and Hastie (2019)).

Limits

1. Maximum number of features: the PDP analysis is limited to 2 features.
2. Feature distribution: some PDP do not show the feature distribution and it can be misleading because you might overinterpret regions with almost no data.
3. Heterogeneous effects might be hidden because PD plots only show the average marginal effects. Solution: Individual Conditional Expectation (ICE) curves.
4. Assumption of independence is the biggest issue with PD plots.

Individual Conditional Expectation (ICE)

Individual Conditional Expectation (ICE) plots display one line per instance that shows how the instance's prediction changes when a feature changes.

Definition

For each instance, $\left\{x_s^{(i)}, x_c^{(i)}\right\}$ the Individual Conditional Expectation (ICE) curve associated to the feature X_s corresponds to the model outcome $\hat{f}$ plotted against the values $x_s \in \Omega_s$ of X_s while $x_c^{(i)}$ remains fixed.

$$ICE_{s,i}(x_s) = \hat{f}(x_s, x_c^{(i)}) \quad \forall x_s \in \Omega_s$$

Individual Conditional Expectation (ICE)

It can be hard to tell whether the ICE ether the ICE curves differ between individuals because they start at different predictions. A simple solution is to center the curves at a certain point in the feature and display only the difference in the prediction to this point.

Definition

The centered ICE is defined as:

$$ICE_{s,i}^{cent}(x_s) = \hat{f}(x_s, x_c^{(i)}) - \hat{f}(x_a, x_c^{(i)}) \quad \forall x_s \in \Omega_s$$

where x_a is the anchor point.

Advantages

1. Intuitive: ICE curves are even more intuitive to understand than PDP.
2. Heterogeneous effects: unlike PDP, ICE curves can uncover heterogeneous relationships.

Limits

1. Maximum number of features: ICE curves can only display one feature meaningfully.
2. Assumption of independence: ICE curves suffer from the same problem as PDPs: If the feature of interest is correlated with the other features, then some points in the lines might be invalid data points according to the joint feature distribution
3. Readibility: If many ICE curves are drawn, the plot can become overcrowded.

Accumulated Local Effects (ALE)

Definition

Accumulated Local Effects (ALE) plots describe how features influence the prediction of a ML model on average, while taking into account the dependence between the features.

1. ALE averages the changes of predictions (defined as the gradient), not the predictions itself. The change is defined as the gradient.
2. ALE accumulates the local gradients over the range of features in set Ω_s, which gives us the effect of the feature on the prediction.
3. The derivative component of the ALE isolates the effect of the feature of interest and blocks the effect of correlated features.

Accumulated Local Effects (ALE)

Definition

The (uncentered) ALE average the changes in the predictions and accumulate them over an interval on X_s values

$$ALE_s(x) = \int_{z_{0,s}}^{x} \mathrm{E}_{X_c|X_s}\left(\frac{\partial \hat{f}\left(X_s, X_C\right)}{\partial X_s}|X_s = z_s\right) \qquad \forall x_s \in \Omega_s$$

where $z_{0,s}$ is the minimum value of X_s for which the ALE curve is computed.

Definition

The centered ALE is defined as:

$$ALE_s^{cent}(x) = ALE_s(x) - \mathrm{E}_{X_s}\left(ALE_s(X_s)\right)$$

Advantages

1. Unbiased: which means that ALE plots still work when features are correlated. PDP fail in this scenario because they marginalize over unlikely or even physically impossible combinations of feature values.
2. Faster to compute than PDPs, since the largest possible number of intervals is the number of instances with one interval per instance. The PDP requires n times the number of grid points estimations. For k grid points, PDPs require k times more predictions than the worst case ALE plot where as many intervals as instances are used.
3. Interpretation is clear. Conditional on a given value, the relative effect of changing the feature on the prediction can be read from the ALE plot. ALE plots are centered at zero. This makes their interpretation nice, because the value at each point of the ALE curve is the difference to the mean prediction.

Limits

1. ALE plots can become a bit shaky (many small ups and downs) with a high number of intervals.
2. No perfect solution for setting the number of intervals: If the number is too small, the ALE plots might not be very accurate. If the number is too high, the curve can become shaky.
3. Interpretation remains difficult when features are strongly correlated. Because if they have a very strong correlation, it only makes sense to analyze the effect of changing both features together and not in isolation.
4. Second-order ALE estimates have a varying stability across the feature space, which is not visualized in any way. The reason for this is that each estimation of a local effect in a cell uses a different number of data instances.

Global surrogate models

Definition

A global surrogate model is an interpretable model that is trained to approximate the predictions of a black box model.

How to build a global surrogate model?

1. For the selected dataset X, get the predictions of the black box model.
2. Select an interpretable model type (linear model, decision tree, ...).
3. Train the interpretable model on the dataset X and its predictions.
4. Measure how well the surrogate model replicates the predictions of the black box model (R2, AUC, other metrics). Interpret the surrogate model.

Local surrogate models

Definition

Local surrogate models are interpretable models that are used to explain individual predictions of black box machine learning models.

Various approaches of local surrogate models have been proposed, among which:

1. Local Interpretable Model-agnostic Explanations (LIME) method proposed by Ribeiro et al. (2016)
2. LOcal Rule-based Explanations (LORE) proposed by Guidotti et al. (2019).

Local interpretable model-agnostic explanations (LIME)

The goal is to understand why the machine learning model made a certain prediction.

1. LIME tests what happens to the predictions when you give variations of your data into the machine learning model.
2. LIME generates a new dataset X_{hx} consisting of permuted samples and the corresponding predictions of the black box model.
3. On this new dataset LIME then trains an interpretable model, which is weighted by the proximity of the sampled instances to the instance of interest.
4. The learned model should be a good approximation of the machine learning model predictions locally, but it does not have to be a good global approximation.
5. This kind of accuracy is also called local fidelity.

Advantages

1. Resulting explanations are short (= selective) and possibly contrastive. Therefore, they make human-friendly explanations.
2. Fidelity measure (how well the interpretable model approximates the black box predictions) gives a good idea of how reliable the interpretable model is in explaining the black box predictions in the neighborhood of the data instance of interest.

Shapley Values

Definition

Shapley values – method from coalitional game theory – tells us how to fairly distribute the "payout" among the features.

1. The Shapley value is a solution concept in cooperative game theory.
2. It was named in honor of Lloyd Shapley, who introduced it in 1953 and won the Nobel Prize in Economics for it in 2012.

Remark Shapley values correspond to the contribution of each feature towards pushing the prediction away from the expected value. Thus, the Shapley value for each feature is basically trying to find the correct weight such that the sum of all Shapley values is the difference between the predictions and average value of the model. **How much has each feature value contributed to the prediction compared to the average prediction?**

Advantages

1. Fair distribution: The difference between the prediction and the average prediction is fairly distributed among the feature values of the instance Â–the Efficiency property of Shapley values. This property distinguishes the Shapley value from other methods such as LIME. LIME does not guarantee that the prediction is fairly distributed among the features.
2. Solid theory. The Shapley value is the only explanation method with a solid theory. The axioms – efficiency, symmetry, dummy, additivity – give the explanation a reasonable foundation.

Limits (1/2)

1. Computing time: In 99.9% of real-world problems, only the approximate solution is feasible. An exact computation of the Shapley value is computationally expensive because there are 2^k possible coalitions of the feature values and the "absence" of a feature has to be simulated by drawing random instances, which increases the variance for the estimate of the Shapley values estimation.
2. The exponential number of the coalitions is dealt with by sampling coalitions and limiting the number of iterations M. Decreasing M reduces computation time, but increases the variance of the Shapley value.

Limits (2/2)

1. Interpretation: The Shapley value can be misinterpreted. The Shapley value of a feature value is not the difference of the predicted value after removing the feature from the model training.
2. Independence assumption: the Shapley value method suffers from inclusion of unrealistic data instances when features are correlated. To simulate that a feature value is missing from a coalition, we marginalize the feature. This is achieved by sampling values from the feature's marginal distribution. This is fine as long as the features are independent.

SHapley Additive exPlanations (SHAP)

Definition

SHAP (SHapley Additive exPlanations) by Lundberg and Lee (2016) is a method to explain individual predictions. SHAP is based on Shapley Values. The goal of SHAP is to explain the prediction of an instance x by computing the contribution of each feature to the prediction.

1. SHAP only consider a subset M of coalitions randomly drawn from the set of all coalitions,
2. One innovation of SHAP is that the Shapley value explanation is represented as an additive feature attribution method, a linear model.

Notation

- $z \in \{0,1\}^M$ is a coalition vector, i.e. an entry of 1 means that the corresponding feature value is "present" and 0 that it is "absent".
- M denotes the maximum coalition size.
- $\phi_j\left(\hat{f}\right) = \phi_j \in \mathbb{R}$ is the Shapley value for a feature j.

Definition

The SHapley Additive exPlanations (SHAP) model $g(.)$ is defined as:

$$\hat{f}(z') = \phi_0 + \sum_{j=1}^{M} \phi_j \; z_j$$

where $\phi_0 = \sum_{i=1}^{n} \hat{f}(x_i)$ is the base value defined as the average of individual predicted outcomes.

Interpretation (1/2)

- You can visualize feature attributions such as Shapley values as "forces".
- Each feature value is a force that either increases or decreases the prediction.
- $\phi_j\left(\hat{f}\right) = \phi_j \in \mathbb{R}$ is the Shapley value for a feature j.

Definition

The SHAP feature importance is defined as:

$$I_j = \sum_{i=1}^{n} |\phi_j^{(i)}|$$

The idea behind SHAP feature importance is simple: Features with large absolute Shapley values are important.

Interpretation (2/2)

Definition

The **SHAP summary plot** combines feature importance with feature effects. Each point on the summary plot is a Shapley value for a feature and an instance. The position on the y-axis is determined by the feature and on the x-axis by the Shapley value. The color represents the value of the feature from low to high.

Definition

The **SHAP dependence plot** simply displays the scatter plot of the feature value and the associated Shapley values for all the instances.

Definition

SHAP clustering works by clustering on Shapley values of each instance. This means that you cluster instances by explanation similarity.